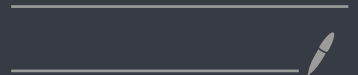


IEEE 754

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1) $0.1\ 000\ 0000.01\ 0000\ 0000\ 0000\ 0000\ 0 = 2,5$

VZ
CH
Mantisse

1 = -
0 = +

$$CH = 1000\ 0000_2 \hat{=} 128_{10}$$

$$E = CH - 127 = 128 - 127 = 1$$

$$M = 1,01$$

$$\Rightarrow Z = 2^1 \cdot 1,01$$

$$= 10,1_2 = 2,5_{10}$$

$$2 + 2^1 = 2,5$$

2) $1,100\ 000\ 10.10101101\ 00\ 000000\ 0000000 = -13,40625_{10}$

VZ = -
CH

$$CH = 1000\ 0010 \hat{=} 130$$

$$128 \quad 2$$

$$E = 130 - 127 = 3$$

$$M = 1,1010\ 1101$$

$$\Rightarrow Z = 2^3 \cdot (1,1010\ 1101)$$

$$= 1101,01101$$

$$8 \quad 4 \quad 1 \quad 0,25 \quad 0,125 \rightarrow 0,03125$$

$$= -13,40625_{10}$$

$$3) \quad 0. \underbrace{1000 \ 1110}_{CH=142} . 1100 \ 00100 \ 11 \ 0000 \ 0100000 \ 1 - 57648, 25390625$$

$$E = 142 - 127 = 15$$

$$M = 1,1100 \ 00100 \ 11 \ 0000 \ 01000001$$

$$Z = 2^{15} \cdot (1,1100 \ 00100110000 \ 01000001)$$

$$Z = 11100 \ 00100110000,01000001 = 57648, 25390625$$

32768 (from 11100)
 16384 (from 00100)
 8192 (from 110000)
 $256 + 32 + 16$ (from 0010011)
 $2^2 = 0,25$ (from 01)
 $2^{-8} = 0,00390625$ (from 000001)

$$4) \quad 14592,0625_{10} =$$

$$\begin{aligned} \cdot 14592 : 2 &= 7296 \text{ OR} \\ 7296 : 2 &= 3648 \text{ OR} \\ 3648 : 2 &= 1824 \text{ OR} \\ 1824 : 2 &= 912 \text{ OR} \\ 912 : 2 &= 456 \text{ OR} \\ 456 : 2 &= 228 \text{ OR} \\ 228 : 2 &= 114 \text{ OR} \\ 114 : 2 &= 57 \text{ OR} \\ 57 : 2 &= 28 \text{ OR} \\ 28 : 2 &= 14 \text{ OR} \\ 14 : 2 &= 7 \text{ OR} \\ 7 : 2 &= 3 \text{ OR} \\ 3 : 2 &= 1 \text{ OR} \\ 1 : 2 &= 0 \text{ OR} \end{aligned}$$

$$\begin{aligned} \cdot 0,0625 \cdot 2 &= 0,125 \text{ 0} \\ 0,125 \cdot 2 &= 0,250 \text{ 0} \\ 0,250 \cdot 2 &= 0,5 \text{ 0} \\ 0,5 \cdot 2 &= 1 \text{ 1} \end{aligned}$$

$$0,0001$$

$$\Rightarrow 11 \ 10010000 \ 0000,0001 = 1,110010000 \ 0000 \ 0001 \cdot 10_2^{1101}$$

$$E = e + B \quad (B = 127_{10})$$

$$E = 13_{10} + 127_{10}$$

$$E = 140_{10}$$

$$E = 1000 \ 1100_2$$

$$\begin{aligned} 140 : 2 &= 70 \text{ OR} \\ 70 : 2 &= 35 \text{ OR} \\ 35 : 2 &= 17 \text{ OR} \\ 17 : 2 &= 8 \text{ OR} \\ 8 : 2 &= 4 \text{ OR} \\ 4 : 2 &= 2 \text{ OR} \\ 2 : 2 &= 1 \text{ OR} \\ 1 : 2 &= 0 \text{ OR} \end{aligned}$$

$$= 0,10001100 \cdot 1100 \ 1000 \ 0000 \ 0000 \ 1 \ 000000$$

V_2 Exponent

Mantissa

$$5) - 2129,46875 =$$

$$\begin{array}{lcl} 2129 : 2 = 1064 & 1R \\ 1064 : 2 = 532 & 0R \\ 532 : 2 = 266 & 0R \\ 266 : 2 = 133 & 0R \\ 133 : 2 = 66 & 1R \\ 66 : 2 = 33 & 0R \\ 33 : 2 = 16 & 1R \\ 16 : 2 = 8 & 0R \\ 8 : 2 = 4 & 0R \\ 4 : 2 = 2 & 0R \\ 2 : 2 = 1 & 0R \\ 1 : 2 = 0 & 1R \end{array}$$

$$\begin{array}{lcl} 0,46875 \cdot 2 = 0,9375 & 0 \\ 0,9375 \cdot 2 = 1,875 & 1 \\ 0,875 \cdot 2 = 1,75 & 1 \\ 0,75 \cdot 2 = 1,5 & 1 \\ 0,5 \cdot 2 = 1 & 1 \end{array}$$

0,011110

$$\Rightarrow 1000 \ 0101 \ 0001,01111_2 = 1,000 \ 0101 \ 0001 \ 01111 \cdot 10_2^{10111}$$

$$E = C + B \quad (B = 127_{10})$$

$$E = 11_{10} + 127_{10}$$

$$E = 138_{10}$$

$$E = 1000 \ 1010$$

$$\begin{array}{lcl} 138 : 2 = 69 & 0R \\ 69 : 2 = 34 & 1R \\ 34 : 2 = 17 & 0R \\ 17 : 2 = 8 & 1R \\ 8 : 2 = 4 & 0R \\ 4 : 2 = 2 & 0R \\ 2 : 2 = 1 & 0R \\ 1 : 2 = 0 & 1R \end{array}$$

$$\Rightarrow 1,1000 \ 1010 \ 000 \ 0101 \ 0001 \ 01111 \ 0000000$$

$$6) - 564, 275625 \cdot 765625 =$$

$$\begin{array}{l} 564 : 2 = 282 \text{ OR} \\ 282 : 2 = 141 \text{ OR} \\ 141 : 2 = 70 \text{ 1R} \\ 70 : 2 = 35 \text{ OR} \\ 35 : 2 = 17 \text{ 1R} \\ 17 : 2 = 8 \text{ 1R} \\ 8 : 2 = 4 \text{ OR} \\ 4 : 2 = 2 \text{ OR} \\ 2 : 2 = 1 \text{ OR} \\ 1 : 2 = 0 \text{ 1R} \end{array}$$

$$\begin{array}{l} 0,275625 \cdot 2 = 0,55125 \text{ 0} \\ 0,55125 \cdot 2 = 1,1025 \text{ 1} \\ 0,1025 \cdot 2 = 0,205 \text{ 0} \\ 0,205 \cdot 2 = 0,41 \text{ 0} \\ 0,41 \cdot 2 = 0,82 \text{ 0} \\ 0,82 \cdot 2 = 1,64 \text{ 1} \\ 0,64 \cdot 2 = 1,28 \text{ 1} \\ 0,28 \cdot 2 = 0,56 \text{ 0} \\ 0,56 \cdot 2 = 1,12 \text{ 1} \\ 0,12 \cdot 2 = 0,25 \text{ 0} \\ 0,25 \cdot 2 = 0,5 \text{ 0} \\ 0,5 \cdot 2 = 1 \text{ 1} \end{array}$$

$$\Rightarrow 10 \text{ } 00110100, 010001101001_2 = 1,000110100010001101001_2$$

$$E = 2_{10}$$

$$E = 9_{10} + 127_{10}$$

$$E = 136_{10}$$

$$E = 10001000$$

$$\begin{array}{l} 136 : 2 = 68 \text{ OR} \\ 68 : 2 = 34 \text{ OR} \\ 34 : 2 = 17 \text{ OR} \\ 17 : 2 = 8 \text{ 1R} \\ 8 : 2 = 4 \text{ OR} \\ 4 : 2 = 2 \text{ OR} \\ 2 : 2 = 1 \text{ OR} \\ 1 : 2 = 0 \text{ 1R} \end{array}$$

$$\Rightarrow 1,10001000, 00011010001000110100100$$